

Event- and Weather-Aware Spatio-Temporal Graph Networks for Traffic Shockwave Forecasting

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Abstract

Short-term traffic forecasts feed operational decisions such as signal control, detour guidance, and transit dispatch. Their value is highest during unusual periods, for example heavy rain or the departure peak after a stadium event, yet standard benchmarks average errors over test sets dominated by routine traffic. We ask whether weather and scheduled-event information, both available before congestion forms, improve speed forecasts specifically inside such anomalous windows. On the PEMS-BAY network of 325 sensors we condition a Spatio-Temporal Graph Convolutional Network on calendar, weather, event-activity, event-magnitude, and venue-distance channels, and evaluate it inside windows defined only from schedules and weather observations, alongside a physics-level check of whether the model reproduces the upstream propagation of a shockwave. Across 81 trained configurations, aggregate MAE and decision-relevant metrics disagree almost completely: their rank correlation with propagation fidelity is -0.23 ($p = 0.55$). Weather channels lower aggregate error slightly while making the model worse inside both weather windows; a congestion-weighted loss leaves aggregate MAE unchanged while raising onset recall by four points; and a directed diffusion operator, adopted precisely because shockwaves travel upstream $1.68\times$ more often than downstream, leaves aggregate MAE inside seed noise while measurably reducing the model’s reproduction of that asymmetry. What consistently helps is spatial structure in the input rather than more context or a more expressive operator: two configurations with identical parameter and channel counts reverse their ordering on every window and every fold according to whether their added channels vary across nodes.

1 Introduction

Traffic forecasts are rarely an end in themselves. Agencies use them to time signal plans, position staff, reroute busses, and warn drivers. These interventions matter most when traffic breaks from its daily pattern, for example, while rain reduces road capacity or while 17,000 hockey fans leave a downtown arena within half an hour. Such periods make up a small part of any test set. A model can therefore reduce its average error on a standard benchmark without becoming more useful in the situations where a forecast actually

changes a decision.

Spatio-temporal graph neural networks are the established tool for this forecasting task. Sensors become graph nodes, road relationships become edges, and recent measurements become node-level time series. STGCN (Yu, Yin, and Zhu 2018) and DCRNN (Li et al. 2018) showed that this structure predicts road speeds well from historical observations alone. Later work added learned graph structure (Wu et al. 2019; Bai et al. 2020) or contextual attributes such as observed incidents (Fan et al. 2026). However, An incident marker only becomes available once the disruption has started. Scheduled events and weather differ in one operationally useful way that they can be known before congestion forms.

This paper addresses two questions. First, does conditioning an STGCN on weather and scheduled-event information improve accuracy inside exogenously defined anomalous windows, rather than only in the aggregate? Second, is the error profile over distance from the venue and over congestion-onset time consistent with a shockwave that spreads outward through the road graph?

Three difficulties shape the design. Anomalous windows are rare, so aggregate metrics can bury them. Hourly weather records and irregular event schedules must be aligned to a 5-minute sensor grid without leaking future information. And an event flag broadcast identically to all 325 sensors says nothing about where the disruption originates, which is why the distance-to-venue channel carries the spatial anchor.

The intended contributions are a set of anomaly windows defined from schedules and weather rather than from the traffic signal itself, a conditioned version of an established architecture instead of a new one, and an evaluation that adds venue-distance profiles, onset timing, and an operational decision rule to the usual point-error metrics.

2 Related Work

2.1 Forecasting anomalous traffic

Most traffic benchmarks report errors averaged over all test timesteps. Lee et al. (2021) compared major deep forecasting models under consistent settings and found that rankings change in intervals with abrupt speed changes. Zhang et al. (2021) evaluated hybrid models separately under normal conditions, holidays, and adverse weather. Zhang et al. (2023) targeted extreme conditions by retrieving similar

historical patterns at test time, and Kralj et al. (2025) proposed a sudden-event accuracy measure after showing that average error can stay strong while sharp slowdowns are missed. In all four studies the difficult intervals are identified from the realized traffic itself. We instead define them from event schedules and weather, which are known independently of the forecast target.

2.2 Spatio-temporal graph forecasting

STGCN combines Chebyshev graph convolution with gated temporal convolution (Yu, Yin, and Zhu 2018); DCRNN places diffusion convolution inside a recurrent encoder-decoder (Li et al. 2018). Graph WaveNet adds a self-adaptive adjacency matrix and dilated temporal convolutions (Wu et al. 2019), and AGCRN learns node-specific parameters without a supplied graph (Bai et al. 2020). Since our contribution is the conditioning and the evaluation protocol, we keep STGCN unchanged as the forecasting backbone rather than adding another architecture.

2.3 External context in graph models

AST-GCN concatenates weather and points of interest to the traffic input (Zhu et al. 2021), and WeaGAN learns weather-traffic interactions through attention gates (Wang and Luo 2023). On the event side, Li et al. (2023) model congestion episodes as events in a deployed system, Han et al. (2024) fuse traffic with textual event descriptions around convention centers, and GAT-Informer represents a sports event through temporal profiles around its start and end plus each node’s distance from the arena (Song et al. 2024), which makes it the closest prior work. IGSTGNN models the spatial footprint and temporal decay of incidents (Fan et al. 2026). Dynamic graphs, exogenous context, and abnormal-window evaluation have thus each been studied on their own. What is missing, and what we attempt here, is a single controlled pipeline that combines structured event variables including attendance, weather, disruption-window reporting, and distance-based propagation analysis on a standard benchmark. We do not claim the first event-aware or weather-aware traffic model.

3 Methodology

3.1 Problem definition

Let the sensor network be a graph $G = (V, E, A)$ with $|V| = N = 325$ sensors and a weighted adjacency matrix $A \in \mathbb{R}^{N \times N}$. At timestep t the node features are $X_t \in \mathbb{R}^{N \times C}$. Given the past $L = 12$ observations (60 minutes at 5-minute resolution), the task is to predict the next $H = 12$ speed states:

$$\hat{Y}_{t+1:t+H} = f_{\theta}(X_{t-L+1:t}, A), \quad (1)$$

with $X_{t-L+1:t} \in \mathbb{R}^{L \times N \times C}$ and $\hat{Y}_{t+1:t+H} \in \mathbb{R}^{H \times N}$. Speed is the only prediction target for every configuration; the ablation varies only C , which ranges from 1 (speed only) to 11 (all context channels). Errors are reported at horizon steps 3, 6 and 12, i.e. 15, 30 and 60 minutes, matching the convention under which PEMS-BAY results are published (Wu et al. 2019); the 45-minute column is included in Table 6 and changes no conclusion.

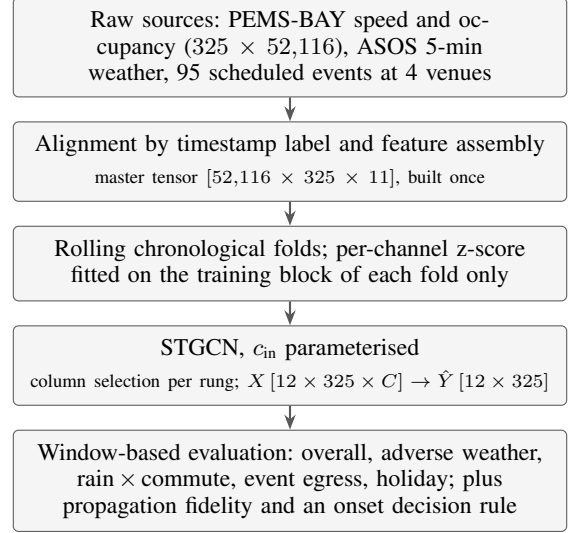


Figure 1: Data and evaluation pipeline. Every stage reads only information available at forecast time; the fold boundaries and the normalization statistics never see validation or test data.

3.2 Pipeline overview

Figure 1 summarizes the pipeline. Raw traffic, weather, and event records are aligned to one 5-minute index, assembled once into an eleven-channel master tensor, split into rolling chronological folds, and normalized per channel with training statistics only. Every ablation rung selects *columns* of that one master tensor rather than rebuilding it, so all rungs share the same samples, the same fold boundaries and the same scaler, and their errors remain directly comparable.

3.3 Graph construction and samples

Sensor identifiers, the adjacency matrix and pairwise road distances come from the Augmented PEMS-BAY release (george-j-ste 2024; Li et al. 2018). A recorded driving distance d_{ij} becomes a Gaussian edge weight, and weak connections are removed:

$$A_{ij} = \begin{cases} \exp \left[- (d_{ij} / \sigma_d)^2 \right] & \text{if this value} \geq 0.1, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where σ_d is the standard deviation of the finite recorded distances.

A is *directed*: it encodes driving distance, and 1,789 of its edges have no reverse. This matters twice, in opposite directions. Chebyshev graph convolution is defined on an undirected graph and needs a symmetric Laplacian with a real spectrum, so the backbone symmetrizes with $\max(A, A^T)$ before computing the Laplacian. Passing the raw directed matrix to a symmetric eigensolver instead returns $\lambda_{\max} = 1.267$ against a true largest eigenvalue of 1.001, mis-scaling the Chebyshev basis by 27%. The *analysis* of Section 3.9, by contrast, requires travel direction and reads the original directed matrix.

The time axis is taken from the column labels of the speed file, never regenerated: local wall-clock time in the study period is missing 2017-03-12 02:00–02:55 to daylight saving, so a naive 5-minute range yields $181 \times 288 = 52,128$ stamps against the 52,116 the data actually contains, and every later timestamp would slide by one hour. All sources are joined to that axis *by label*.

3.4 Backbone

The backbone follows Yu, Yin, and Zhu (2018). Two spatio-temporal blocks each apply a gated temporal convolution, a graph convolution, a second gated temporal convolution, layer normalization, dropout of 0.3, and a residual connection; both temporal convolutions use kernel size $K_t = 3$, and channel widths are $[c_{\text{in}}, 64, 64]$ and $[64, 64, 64]$. An output convolution over the remaining time axis and a linear layer emit all 12 horizons in one pass. The speed-only model has 116,940 trainable parameters and the eleven-channel model 121,420.

The default graph operator is the Chebyshev approximation of order $K_s = 3$,

$$\Theta *_G x = \sum_{k=0}^{K_s-1} \theta_k T_k(\tilde{L}) x, \quad \tilde{L} = \frac{2}{\lambda_{\max}} L - I_N, \quad (3)$$

with $L = I_N - D^{-1/2} \tilde{A} D^{-1/2}$ the normalized Laplacian of the symmetrized adjacency $\tilde{A} = \max(A, A^T)$ and T_k the Chebyshev polynomials. The graph convolution is general in the input width, $y_j = \sum_{i=1}^{C_{\text{in}}} \Theta_{i,j}(L) x_i$; Yu et al. instantiate it only at $C_{\text{in}} = 1$, and our ablation is precisely a sweep over C_{in} .

We deliberately depart from the reference implementation in three places, and record them because they affect comparability rather than novelty. First, the model emits all H horizons from a single forward pass, where the original predicts one step and recurses at test time; ours is the convention under which the PEMS-BAY reference numbers were published (Wu et al. 2019). Second, the block widths are 64/64/64 rather than the paper’s bottleneck 64/16/64, matching the configuration used for our reproduction. Third, dropout and weight decay are actually active here, whereas the released code collects a weight-decay term it never adds to the loss and feeds `keep_prob=1.0` during training.

3.5 Exogenous conditioning

The conditioned model widens the input from one to eleven channels and changes nothing else. Table 1 lists them together with the axis each varies over, which is the property that turns out to matter most.

Event channels. Each venue e contributes an exponential spatial kernel over great-circle distance $d_{n,e}$ from sensor n ,

$$w_{n,e} = \exp(-d_{n,e}/\tau), \quad \tau = 4 \text{ km}. \quad (4)$$

#	Channel	Group	Varies over
0	Speed (target)	speed	node $\times$ time
1	Occupancy	occupancy	node $\times$ time
2	Time of day	calendar	time
3	Day of week	calendar	time
4	Is holiday	calendar	time
5	Temperature	weather	time
6	Precipitation	weather	time
7	Wind speed	weather	time
8	Distance to venue	event_geo	node
9	Event active decay	event_geo	node $\times$ time
10	Event load	event_att	node $\times$ time

Table 1: The eleven input channels. Calendar and weather channels are broadcast identically to all 325 nodes; the three event channels vary across nodes through a distance kernel. Channels 2–4 pass through unnormalized; the remaining eight are z-scored.

Writing W_e for the active interval of fixture e and A_e for its expected attendance, the two temporal event channels are

$$\text{active}_{n,t} = \sum_e \mathbf{1}[t \in W_e] w_{n,e}, \quad (5)$$

$$\text{load}_{n,t} = \sum_e \mathbf{1}[t \in W_e] \mathbf{1}[A_e \geq A_{\min}] \frac{A_e}{A_{\text{ref}}} w_{n,e}, \quad (6)$$

with $A_{\text{ref}} = 20,000$ and $A_{\min} = 15,000$. The two differ only in whether attendance enters, which makes the pair `4_event_geo` versus `5_event_geo_att` a controlled test of whether crowd size carries signal beyond mere fixture presence. The threshold is measured, not conventional: below 15,000 attendees the egress-hour speed anomaly is $+0.35 \pm 0.41$ mph, i.e. the wrong sign, while the thresholded weight reaches $t = -9.76$.

3.6 Congestion-weighted loss

Aggregate L_1 on speed is dominated by free flow: 90.4% of observations exceed 55 mph and carry 63.7% of the total absolute error at an MAE of 2.18, while the <45 mph regime is 6% of observations carrying 26% of the error at an MAE of 12–14. A loss averaged over all cells therefore spends most of its gradient on predicting 65 mph as 63. As a second, deliberately minimal intervention we reweight the same L_1 towards low speeds,

$$\omega(y) = 1 + \max\left(0, \frac{v_0 - y}{\Delta}\right), \quad (7)$$

$$\mathcal{L} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \omega(y_i) |\hat{y}_i - y_i|, \quad (8)$$

with $v_0 = 60$ mph and $\Delta = 20$ mph, so $\omega = 1.0$ at free flow, 1.5 at 50 mph and 2.5 at 30 mph. The weighting touches the objective only: architecture, channels, splits and reported metrics are unchanged, and every reported number remains *unweighted* MAE, RMSE or MAPE on de-normalized mph. This is the loss-function form of the paper’s own argument: if aggregate MAE hides the intervals that matter, the training objective should not be governed by it either.

3.7 Directed graph operator

Chebyshev convolution cannot distinguish an upstream neighbour from a downstream one, because symmetrization discards exactly the direction in which a shockwave travels. As a third intervention we replace it with the diffusion convolution of Li et al. (2018), a dual random walk over the *directed* adjacency,

$$Z = \sum_{k=0}^K \left(P_f^k X \Theta_{k,1} + P_b^k X \Theta_{k,2} \right), \quad (9)$$

with forward and backward transition matrices $P_f = D_o^{-1}A$ and $P_b = D_i^{-1}A^\top$ on the raw directed adjacency, kept separate so the model learns a different weight upstream and downstream. Nothing else changes: the layer computes $\sum_k S_k x W_k$ and is indifferent to what the supports S_k are, so only their count moves, from K_s to $1 + 2(K_s - 1)$ — at $K_s = 3$ the supports are $[I, P_f, P_f^2, P_b, P_b^2]$, five instead of three, which adds exactly 16,384 parameters ($4 \times 64 \times 64$) in both the speed-only and the eleven-channel model.

3.8 Rolling evaluation and window definitions

Why not a single chronological split. California’s wet season is January to April, so *any* chronological 70/10/20 split of this period puts every rain episode in the training block. We verified this: the single-split test block contains **zero** adverse-weather episodes. A weather-conditioned model cannot differ from a traffic-only one there, whatever it has learned. We therefore use expanding-window rolling folds with a 28-day test block, a 7-day validation block and a minimum 28-day training block.

Fold selection. Folds are selected by the exogenous content of their test block, which is known from schedules and weather records *before* any model is trained. Folds 3–5 contain zero adverse-weather episodes and zero NHL fixtures and cannot inform either research question, so all results use folds 00–02, which between them hold all 51 rain episodes and all 19 NHL fixtures that reach any test block.

Windows. All four evaluation windows are derived from a weather observation or a published schedule, never from speed or occupancy; the endogenous alternative is only available after congestion has already formed, and is the definition this work exists to contrast against. *Adverse weather* is precipitation ≥ 0.51 mm, the 75th percentile of wet steps (-2.22 mph mean anomaly). *Rain $\times$ commute* intersects it with weekdays 07–10 and 15–19. *Event egress* is the hour following the scheduled end of a fixture with $A_e \geq 15,000$. *Holiday* is the five US federal holidays in the period.

Episodes, not timesteps. Neighbouring 5-minute steps and neighbouring sensors are strongly correlated, so a standard error over node-timesteps understates the true uncertainty by roughly $\sqrt{n_{\text{steps}}/n_{\text{episodes}}}$. We therefore report the number of *episodes* behind every window, and never attach an error bar to a window backed by fewer than about ten.

A confound we do not hide. An in-window versus out-of-window ratio is always confounded, because a window sits at a particular hour and season rather than being a random sample. Two of ours show it plainly. Event egress reads *easier* than its complement, not because fixtures make traffic predictable but because egress falls at 21:30–23:00 on an empty network. Rain $\times$ commute reads $1.75\text{--}2.93\times$ its complement, which invites “rain triples the error”, but decomposing it 2×2 shows time of day alone accounts for a factor of 2.55 on dry samples in all three folds, with rain adding between -0.65 and $+1.16$ mph on top and not even agreeing in sign. Windows are the right place to *look*; they do not by themselves *attribute*. Every comparison in this paper is therefore model-against-model inside one window, read down a column.

3.9 Metrics

Point error is reported as MAE, RMSE and masked MAPE on de-normalized speed, the masked variant excluding cells below 1 mph where a percentage error is noise. Those three are standard; the two definitions below are not, and they carry the part of the evaluation this paper adds.

Shockwave label. A cell is in shockwave onset when speed has fallen sharply and is now low:

$$s_{n,t} = \mathbf{1}[v_{n,t-3} - v_{n,t} > 10 \text{ mph}] \cdot \mathbf{1}[v_{n,t} < 45 \text{ mph}]. \quad (10)$$

Propagation fidelity. Let $E_d = \{(i, j) : A_{ij} > 0, i \neq j\}$, so that j is reachable driving from i and hence *downstream* of i . A shockwave travels the other way: j breaks down first and i follows. We count ordered co-occurrences in both directions at lag ℓ and report their ratio,

$$R(\ell) = \frac{\sum_{(i,j) \in E_d} \sum_t s_{j,t} s_{i,t+\ell}}{\sum_{(i,j) \in E_d} \sum_t s_{i,t} s_{j,t+\ell}}. \quad (11)$$

$R = 1$ would mean the label is picking up network-wide congestion rather than a travelling wave, so this doubles as a sanity check on the label itself. Evaluated on the *observed* series it peaks at $R = 1.68$ at a 15-minute lag. Evaluated on a model’s own predictions it asks a question no point-error metric asks: has the model learned a travelling wave, or a smooth conditional mean that happens to have low MAE?

Onset recall and false alarms. A reactive rule acts once congestion is observed: speed below 45 mph for three consecutive steps. The forecast-based policy acts as soon as the model predicts that. Onset recall is the share of observed onsets the model flags; false alarms are flags with no corresponding onset. Because false alarms and misses carry asymmetric costs, and because the two users of this forecast sit at opposite ends of that asymmetry, we sweep the miss-to-false-alarm cost ratio over $\{1, 2, 5, 10, 20, 50\}$ rather than committing to one operating point.

Source	Size	Variables
PEMS-BAY (Li et al. 2018; george-jste 2024)	325 sensors, 52,116 5-min steps	speed, occupancy, sensor metadata, road distances
Weather (Iowa Environmental Mesonet 2024)	2 ASOS stations (SJC, NUQ), 5-min grid	temperature, precipitation, wind speed
Scheduled events	95 fixtures, 4 venues	start/end, venue, coordinates, attendance, type, provenance

Table 2: Data sources for January–June 2017.

4 Experiment

4.1 Data and descriptive statistics

The study covers 1 January to 30 June 2017 in the San Francisco Bay Area. Table 2 lists the three sources.

Traffic. Missing speed values are rare (521 of $52,116 \times 325$ cells) and interpolated linearly in time. The target is heavily skewed towards free flow: 90.4% of observations exceed 55 mph, 3.6% fall in 45–55, 2.1% in 35–45 and 3.9% below 35.

Events. The table holds 27 NHL, 24 AHL, 9 MLS and 35 concert or other fixtures, at SAP Center (69), Shoreline Amphitheatre (12), Avaya Stadium (9) and Levi’s Stadium (5); 67 meet the 15,000-attendee threshold. Venue coordinates were cross-checked against Nominatim and Wikidata. Provenance is carried per row: only 37 fixtures have an attendance figure verified against a published box score, and the remaining 58 carry a per-type default we assigned ourselves. That distinction is not bookkeeping. Stratifying the egress-hour effect by it, the 37 verified fixtures show a speed anomaly of -0.63 ± 0.26 mph and a shockwave odds ratio of $5.25 \times$ ($z = 3.66$), whereas the default-attendance tier shows $+0.18 \pm 0.11$ and no effect. Filtering by an attendance threshold alone would filter our own defaults.

Weather. Thirteen ASOS stations cover the region; SJC and NUQ alone place every sensor within 13.2 km, while the eight outer stations win zero sensors and two nearby ones report no night observations at all. Interpolating from two points is rank-one, which is a real limitation and the reason the weather channels carry no spatial structure. The precipitation field `p01i` is a backward one-hour accumulation forward-filled across the following twelve 5-minute steps, so it *cannot be summed*; thresholds and episode counts are unaffected because they are per-step comparisons.

Fold	Train	Test block	Rain	Events	Hol.
fold00	6,048	01-29–02-26	27	4	1
fold01	14,112	02-26–03-26	12	10	0
fold02	22,141	03-26–04-23	12	8	0
<i>fold03–05</i>	<i>114,807</i>	<i>04-23–06-30</i>	<i>0</i>	<i>5</i>	<i>1</i>
<i>single 70/10/20</i>	<i>36,449</i>	<i>05-25–06-30</i>	<i>0</i>	<i>3</i>	<i>1</i>

Table 3: Rolling folds. “Train” counts samples; rain, events and holidays are *episodes* in the test block. The excluded rows are shown for contrast: folds 3–5 hold 73% of the total training compute and zero rain episodes, and the conventional chronological split holds none either.

Measured effects, before any training. Adverse weather lowers speed by 2.22 mph on average. The egress hour of an evening NHL fixture at SAP Center lowers it by 1.69 ± 0.51 mph within 2 km. Holidays are the *largest* exogenous effect and the sign is positive: Memorial Day runs $+9.50$ mph in the morning peak and $+11.27$ in the evening, because a holiday removes the commute rather than adding a crowd. Holidays are therefore reported as a separate stratum and never merged into “adverse”.

4.2 Preprocessing

Weather and event records are joined to the speed file’s own timestamp labels. Weekends and holidays are *kept*, against Yu et al., who drop them to eliminate atypical traffic: 46 of our 95 fixtures fall on a weekend, and discarding them would discard half the phenomenon. Sliding windows of 12 input and 12 target steps are cut, the folds of Section 3.8 are applied, and each channel is z-scored with statistics from that fold’s training block only; the three calendar channels pass through unnormalized so that a 0/1 holiday flag is not averaged with miles per hour and millimetres. The master tensor is built once and rungs select columns from it.

4.3 Splits

Table 3 gives the three folds and their exogenous content. The imbalance is the point: it is why folds 3–5 are excluded and why the conventional split cannot be used at all.

4.4 Setup

Nine variants are trained on three folds with three seeds each, for 81 runs: five ablation rungs, plus `0_speed` and `6_all` again with the congestion-weighted loss, and again with the diffusion operator. Optimization uses Adam at learning rate 10^{-3} , weight decay 10^{-4} , batch size 64, gradient clipping at 5.0 and a StepLR schedule (step 10, factor 0.7), with early stopping on validation MAE at patience 20; runs stopped between 65 and 100 epochs. Training used one NVIDIA H200 NVL at roughly 3 s per epoch. Two trivial baselines frame the learned models: persistence repeats the last observed speed, and the historical average predicts each sensor’s mean speed for each slot of the week from training data alone.

Horizon	MAE (mph)		RMSE (mph)		MAPE (%)	
	ours	ref.	ours	ref.	ours	ref.
15 min	1.44	1.36	3.06	2.96	3.03	2.90
30 min	1.93	1.81	4.36	4.27	4.35	4.17
60 min	2.58	2.49	5.83	5.69	6.25	5.79

Table 4: Speed-only STGCN on PEMS-BAY under the conventional split: mean over seeds 42–44 against the reference values of Wu et al. (2019). The gap stays within 0.1 mph MAE at every horizon. *Caveat.* These runs predate a correction to the Chebyshev rescaling: `eigsh` assumes a symmetric matrix and does not check, so on the directed adjacency it returned $\lambda_{\max} = 1.2670$ against a true 1.0013 and the spectrum sat in $[-1, 0.58]$. The reproduction has not been repeated since that fix, so it establishes that the loader, sample slicing and training loop are sound, but not the graph operator in its current form.

4.5 Reproduction of the speed-only backbone

Before adding any channel we reproduced vanilla STGCN on plain PEMS-BAY under the conventional split. Table 4 compares three seeds against the STGCN values Wu et al. (2019) report. We note that the PEMS-BAY reference numbers everyone quotes come from that paper, not from Yu et al. (2018), who evaluated on BJER4 and PeMSD7 with a month-based split.

4.6 Main result

Table 5 is the central result. Reading down any single column is meaningful; reading across is the mistake the paper is about.

Five metrics, different winners. The model that wins on aggregate MAE, `5_event_geo_att`, ranks sixth of nine on reproducing shockwave physics and fourth inside the rain $\times$ commute window. The model that best reproduces the physics, `3_weather`, is simultaneously *last* on onset recall and last inside that window. The model with the lowest error where it rains during the commute, `1_traffic`, ranks only fifth on aggregate MAE. The model that catches the most onsets, `6_all` with the weighted loss, ranks third on MAE and carries the most false alarms. The rank correlation between aggregate MAE and propagation fidelity is -0.23 ($p = 0.55$), indistinguishable from none. An aggregate metric does not merely under-state the differences that matter; it *selects a different model* from the one any decision-relevant metric selects.

Weather channels make the model worse. `3_weather` loses to `1_traffic` on every error metric — overall, inside adverse weather, inside rain $\times$ commute, and inside the egress hour — and loses to the plain speed-only baseline inside both weather windows (5.559 against 5.349 on rain $\times$ commute, +3.9%) and on onset recall (0.688, the worst of any trained model). It does rank *first* on propagation fidelity, which is not a contradiction but a further in-

stance of the disagreement above: reproducing the direction in which a wave travels and predicting its magnitude are different abilities, and no point-error metric separates them. The data said why before any training. Controlling for time of day, rain does not change how often breakdowns occur — 91.8% of commute-peak timesteps carry a shockwave in rain against 93.1% dry — nor how large they are (6.25 sensors involved against 6.46). Rain is a uniform capacity reduction of -3.04 mph in the peak, and a uniform slowdown is already legible in the speed history the model has.

Spatial structure, not channel count, decides.

`3_weather` and `5_event_geo_att` have *identical* parameter counts (118,732) and identical channel counts ($C = 5$). They differ only in whether the three added channels vary across nodes, and the outcome reverses in every window and on every fold: inside rain $\times$ commute, 5.559 for the broadcast channels against 4.991 for the spatially varying ones. On a graph model a channel with no spatial variation is not a weak feature but a harmful one: passed through a graph convolution a broadcast scalar is close to a global bias, and it spends three dimensions of capacity saying nothing about which node is which. This also reframes `5_event_geo_att`, which wins in windows containing no events at all, so its gain is unlikely to be event timing; the likely mechanism is `dist_to_venue`, the only channel that distinguishes one sensor from another in a backbone with no node embeddings.

The weighted loss trades false alarms for recall at no cost in MAE.

On `6_all`, reweighting leaves aggregate MAE unchanged ($2.361 \rightarrow 2.356$, against a seed and fold spread of ± 0.25) while raising onset recall from 0.738 to 0.780, at 38% more false alarms; on `0_speed` it costs 3.8% of aggregate MAE for the same 4-point recall gain, at 64% more false alarms. Whether that trade is worth taking depends on the cost ratio, and the two users of this forecast sit at opposite ends of it. A network operator pre-deploying a signal plan pays staff hours for a false alarm and hours of egress delay for a miss, a ratio of 20–50. A navigation service pays one driver a 2–5 minute detour against a measured median 1.3 and p90 5.3 minute delay for driving into the jam, a ratio of 1–3. The operator can afford false alarms but we can evaluate that decision only on 21 positives; the driver cannot afford them and we can evaluate it on 16,743 onsets. The decision with the most tolerance for error is the one this dataset is least able to assess.

4.7 Does the model learn the wave?

Figure 2 answers Section 3.9’s question. Observed shockwave onsets travel upstream $1.68\times$ more often than downstream at a 15-minute lag. Every Chebyshev model reproduces 82–86% of that peak at the 60-minute horizon and none collapses to 1.0, so all are learning a travelling wave rather than a smooth conditional mean — but none reaches the truth either, and the ranking is uncorrelated with MAE.

Model	Overall MAE ↓	Rain×commute MAE ↓	Propagation ↑	Onset recall ↑	False alarms ↓
5_event_geo_att	2.303 (1)	4.991 (4)	1.324 (6)	0.752 (3)	1,823 (7)
6_all + diffusion	2.338 (2)	5.418 (8)	1.315 (7)	0.740 (4)	1,784 (6)
6_all + weighted	2.356 (3)	5.275 (5)	1.376 (4)	0.780 (1)	2,445 (9)
6_all	2.361 (4)	5.283 (6)	1.389 (3)	0.738 (5)	1,775 (5)
1_traffic	2.369 (5)	5.147 (1)	1.452 (2)	0.726 (6)	1,770 (4)
3_weather	2.485 (6)	5.559 (9)	1.454 (1)	0.688 (9)	1,381 (2)
0_speed (published STGCN)	2.502 (7)	5.349 (7)	1.375 (5)	0.702 (7)	1,433 (3)
0_speed + diffusion	2.503 (8)	5.372 (3)	1.246 (9)	0.701 (8)	1,380 (1)
0_speed + weighted	2.596 (9)	5.774 (2)	1.306 (8)	0.747 (4)	2,354 (8)
persistence	3.043	7.183	—	0.654	1,748
historical average	2.776	5.973	—	0.770	4,773

Table 5: Main comparison: mean over 3 seeds × 3 folds at the 60-minute horizon, rank in parentheses. Observed propagation ratio is 1.683. Spearman rank correlation between overall MAE and propagation fidelity is -0.23 ($n = 9$, $p = 0.55$); over the seven Chebyshev variants alone it is -0.11 ($p = 0.82$).

Model	Overall MAE				Window MAE @60 min		
	15	30	45	60	wx	rain×cmt	egr.
5_event_geo_att	1.370	1.789	2.070	2.303	2.534	4.991	1.239
6_all+diff.	1.390	1.820	2.108	2.338	2.708	5.418	1.329
6_all+weight.	1.402	1.832	2.118	2.356	2.671	5.275	1.304
6_all	1.395	1.830	2.124	2.361	2.716	5.283	1.406
1_traffic	1.371	1.810	2.116	2.369	2.576	5.147	1.250
3_weather	1.402	1.868	2.206	2.485	2.747	5.559	1.302
0_speed	1.392	1.865	2.214	2.502	2.743	5.349	1.307
0_speed+diff.	1.390	1.864	2.213	2.503	2.741	5.372	1.304
0_speed+weight.	1.412	1.906	2.280	2.596	2.879	5.774	1.356
persistence	1.568	2.157	2.628	3.043	3.306	7.183	1.400
hist. average	2.776	2.776	2.776	2.776	3.061	5.973	1.826

Table 6: Full error table, MAE in mph. The 45-minute column is included because the error a 60-minute row reports is dominated by free flow: a per-node AR(12) extrapolator is out by 17.56 mph below 35 mph at 60 minutes against 5.35 mph at 15, while its aggregate column moves only from 1.63 to 3.20. No conclusion in this paper changes between the two horizons. Episodes per fold: adverse weather 27/12/12, rain × commute 10/4/2, egress 4/10/8. The egress column reads lower than the overall column for every model because egress falls at 21:30–23:00 on an empty network; compare down a column, never a window against its complement.

A directed operator makes propagation worse. The diffusion operator was adopted precisely because of that $1.68\times$ asymmetry, on the pre-registered reasoning that an operator able to tell upstream from downstream should learn a truer wave. It does the opposite. Propagation fidelity falls from 1.375 to 1.246 for *0_speed* and from 1.389 to 1.315 for *6_all*; paired by (fold, seed) that is -0.129 ± 0.057 ($t = -6.73$, worse in 9 of 9 pairs) and -0.074 ± 0.100 ($t = -2.24$, worse in 7 of 9), and it holds at the 45-minute horizon too. Aggregate MAE meanwhile moves from 2.502 to 2.503 and from 2.361 to 2.338, well inside the ± 0.15 to ± 0.25 spread across seeds and folds, so an aggregate-only

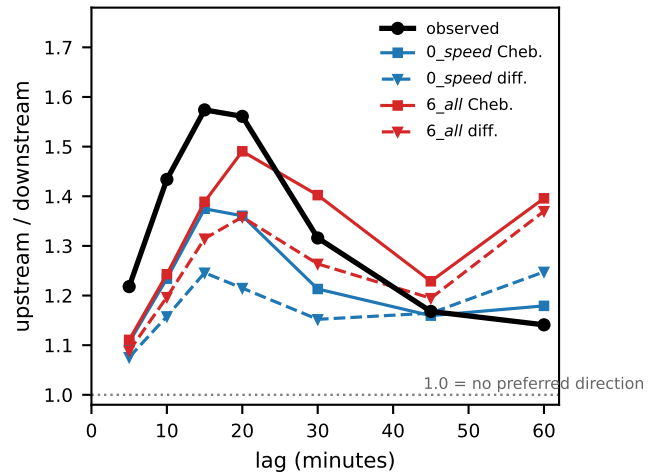


Figure 2: Upstream/downstream co-occurrence of shock-wave onsets against lag, on the observed series and on each model’s own 60-minute predictions (mean of 3 folds × 3 seeds). All models under-reproduce the observed peak at 15 minutes, and the directed diffusion operator under-reproduces it *more* than the symmetric Chebyshev operator it replaces.

report would have logged this as a small improvement while the model moved measurably further from the physics. This is the only single-variable experiment in the set — channels, loss, data and splits are all held fixed — and it sharpens the previous finding: what helps is a channel that tells one node from another, not a mechanism for directional propagation on the graph.

4.8 Negative results worth stating

Compound exposure does not exist here. Adverse precipitation during a qualifying egress hour occurs three times in six months, totalling 18 timesteps, one of them a single 5-minute step. A “rain and event” subset is the natural headline for a paper like this one; on this corpus it would be reporting noise, and we say so rather than constructing it.

Lead time cannot be reported. The natural operational metric — how many minutes earlier than the reactive rule the model triggers — returns a median of *zero* for all nine variants and both trivial baselines, with means below 0.35 minutes. The cause is structural rather than a property of any model: the forecast horizon is 60 minutes and the reactive rule carries its own 15-minute confirmation delay, so the resolvable range is too narrow to separate models. Onset recall and false alarms measure the same operational question and do discriminate, so we report those instead.

Activity without magnitude is expected to be flat. Rung `4_event_geo`, which marks fixture activity without attendance, reaches only $t = -1.02$ against $t = -9.76$ for the attendance-weighted channel. That contrast is a result about the feature, not a failed run.

4.9 Limitations

Two stations cannot express spatial weather structure, and interpolating between them is rank-one; the weather channels are consequently uniform across nodes, which Section 4 shows is the property that makes them harmful here. Venue proximity uses great-circle distance, which does not separate the carriageway that carries egress from the opposing one: 128 sensor pairs lie within 150 m on opposite carriageways, their speeds correlate at 0.115 against 0.850 for same-direction pairs up to 1 km apart, and switching to road-network distance strengthens the measured egress effect from -0.41 ± 0.23 ($t = -1.75$, not significant) to -0.89 ± 0.26 ($t = -3.37$). The event windows are small — 4, 10 and 8 episodes per fold — and the holiday stratum is a single case. Finally, all exogenous channels enter contemporaneously; within the anomalous windows, 23.7% of adverse-weather samples and 15.2% of event samples carry no trace of the exogenous signal anywhere in the 60-minute input window, and a contemporaneous-only model cannot see them at all. Using forecast values instead would be a train/serve mismatch without a 2017 forecast archive to train on.

5 Conclusion

We framed traffic shockwave forecasting as an evaluation problem rather than an architecture problem, and the framing paid off in a way we did not expect. Across 81 trained configurations on PEMS-BAY, aggregate MAE and the metrics an operator would actually use disagree almost completely: their rank correlation with propagation fidelity is -0.23 and not distinguishable from zero, and three separate interventions demonstrate the consequence. Weather channels lower aggregate error slightly while making the model worse inside both weather windows, because rain is a uniform capacity reduction already legible in the speed history rather than a trigger. A congestion-weighted loss leaves aggregate MAE untouched while raising onset recall by four points at 38% more false alarms. And a directed diffusion operator, adopted precisely because shockwaves travel upstream $1.68\times$ more often than downstream, leaves aggregate MAE inside seed noise while measurably *reducing* the

model’s reproduction of that asymmetry — an aggregate-only report would have logged it as an improvement. What consistently helps is not more context or a more expressive operator but spatial structure in the input: two rungs with identical parameter and channel counts reverse their ordering on every window and every fold according to whether their added channels vary across nodes. We therefore recommend that benchmarks in this area report exogenously defined windows, episode counts, and at least one physics-level check alongside point error, and we note as future work that the same protocol should be run where compound exposure actually exists, since on this corpus it amounts to 18 timesteps in six months.

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